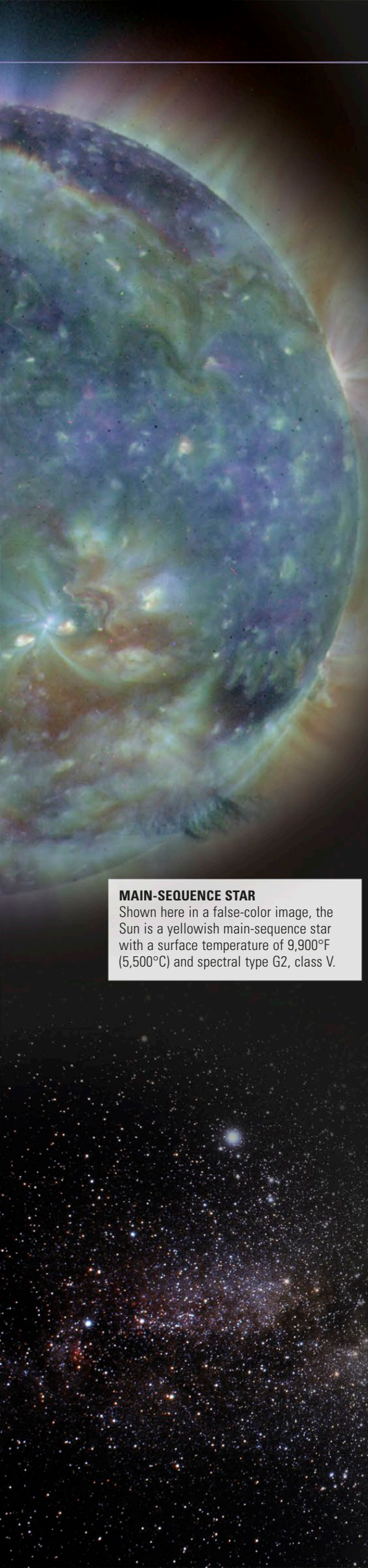
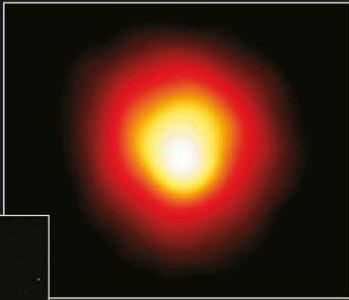




STELLAR SPECTRAL TYPES					
TYPE	PROMINENT SPECTRAL LINES	COLOR		AVERAGE TEMPERATURE	EXAMPLE
O	He ⁺ , He, H, O ²⁺ , N ²⁺ , C ²⁺ , Si ³⁺	Blue		80,000°F (45,000°C)	Gamma Velorum (p.253)
B	He, H, C ⁺ , O ⁺ , N ⁺ , Fe ²⁺ , Mg ²⁺	Bluish white		55,000°F (30,000°C)	Rigel (p.281)
A	H, ionized metals	White		22,000°F (12,000°C)	Sirius (p.252)
F	H, Ca ⁺ , Ti ⁺ , Fe ⁺	Yellowish white		14,000°F (8,000°C)	Procyon (p.284)
G	Ca ⁺ , Fe, Ti, Mg, H, some molecular bands	Yellow		12,000°F (6,500°C)	The Sun (pp.104–107)
K	Ca ⁺ , H, molecular bands	Orange		9,000°F (5,000°C)	Aldebaran (p.256)
M	TiO, Ca, molecular bands	Red		6,500°F (3,500°C)	Betelgeuse (p.256)

STELLAR CLASSIFICATION

Stars are classified by group, according to the characteristics of their spectra. If the light from a star is split into a spectrum, dark absorption and bright emission lines are seen (see p.35). The positions of these lines indicate what elements exist in the photosphere of the star, and the strengths of the lines give an indication of its temperature. The classification system has seven main spectral types, running from the hottest O stars to the coolest M stars. Each spectral type is further divided into 10 subclasses denoted by a number from 0 to 9. Stars are also divided into luminosity classes, denoted by a Roman numeral, which indicates the type of star and its position on the H–R diagram. For example, class V is for main-sequence stars and class II for bright giants, while dim dwarfs are class VI. In addition to the main spectral types, there are classes for stars that show unusual properties, such as the carbon stars (C class). A small letter after the spectral class can also indicate a special property—for example, “v” means variable.



CONTRASTING SUPERGIANTS
Both Betelgeuse (above) and Rigel (left) are supergiants, but they are at opposite ends of the stellar spectrum. Betelgeuse (see p.256) is a cool, red star in its later stages, while Rigel is a hot, blue, relatively young star (see p.281).

MAIN-SEQUENCE STAR
Shown here in a false-color image, the Sun is a yellowish main-sequence star with a surface temperature of 9,900°F (5,500°C) and spectral type G2, class V.

LUMINOSITY

The luminosity of a star is its brightness, defined as the total energy it radiates per second. It can be calculated over all wavelengths—the bolometric luminosity—or at particular wavelengths. Measuring the brightness of a star as it appears in the night sky gives its apparent magnitude, but this does not take account of its distance from Earth. Stars that are located at vastly different distances from Earth can have the same apparent magnitudes if the farther star is sufficiently bright (see p.71). Once a star’s distance is known, its absolute magnitude can be determined. This is its intrinsic brightness, and from this its luminosity can be determined. Stellar luminosities are generally expressed as factors of the Sun’s luminosity. There is a very large range of stellar luminosities, from less than one ten-thousandth to about a million times that of the Sun. If stars are of the same chemical composition, their luminosities are dependent on their mass. Apart from highly evolved stars, they generally obey a consistent mass–luminosity relation, which means that if a star’s luminosity is known, its mass can be determined.

DENEK AND VEGA
Although Deneb (bottom) and its neighbor Vega (top) are similar in apparent brightness, Deneb is about 300 times more distant. If Deneb were moved to Vega’s distance of only 25 light-years from Earth, it would appear to be as bright as a crescent moon.

CECILIA PAYNE-GAPOSCHKIN

Born and educated in England, Cecilia Helena Payne (1900–1979) married fellow astronomer Sergei Gaposchkin. Initially studying at Cambridge University, England, Payne-Gaposchkin was one of the first astronomy graduates to enter Harvard College Observatory. She studied the spectra of stars and suggested in her doctoral thesis that the different strengths of absorption lines in stellar spectra were a result of temperature differences rather than chemical content. She also suggested that hydrogen was the most abundant element in stars.

Her ideas were initially dismissed but finally accepted in 1929.



HARVARD PROFESSOR
Cecilia Payne-Gaposchkin was the first woman to become a full professor at Harvard.